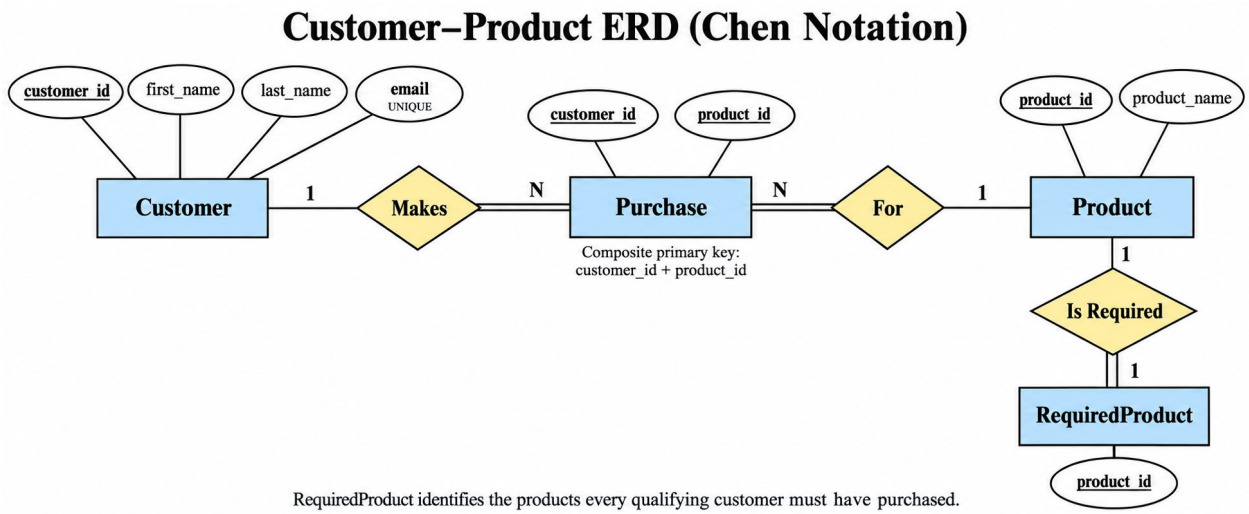
Correlated Subquery

A **correlated subquery** is a query nested inside another query that references a table from an enclosing query. That reference connects the inner query to the row being considered by the outer query.

For example, this query finds customers who have made at least one purchase:



Customer c (customer_id, first_name, last_name, email)
Purchase p (customer_id, product_id)
Product pr (product_id, product_name)
RequiredProduct r (product_id)

```
SELECT c.customer_id, c.first_name, c.last_name
FROM Customer c
WHERE EXISTS (
    SELECT 1
    FROM Purchase p
    WHERE p.customer_id = c.customer_id
);
```

The outer query reads the `Customer` table and assigns it the alias `c`. The inner query reads the `Purchase` table and assigns it the alias `p`.

The connection occurs here:

```
WHERE p.customer_id = c.customer_id
```

Although `p` is defined inside the subquery, **`c` is defined in the outer query**. This reference makes the subquery correlated.

Conceptually, the query checks each customer:

Outer query's current customer	Inner query's question
Customer 1	Does a purchase exist for customer 1?
Customer 2	Does a purchase exist for customer 2?
Customer 3	Does a purchase exist for customer 3?

`EXISTS` returns true when the subquery finds at least one matching row. The outer query therefore includes only customers with purchases. `SELECT 1` supplies a constant value; `EXISTS` checks whether rows are returned, regardless of their values.

The subquery **cannot run independently as written**, because it needs the outer query to define `c`.

By comparison, a **non-correlated subquery** uses only references defined within itself:

```
SELECT customer_id, first_name, last_name
FROM Customer
WHERE customer_id IN (
    SELECT customer_id
    FROM Purchase
);
```

Here, the inner query can run on its own:

```
SELECT customer_id
FROM Purchase;
```

Both examples find customers with purchases, but only the first contains an outer reference. **Correlation depends on that reference, not on the number of tables involved.**